Multi-Machine Monitoring Solution including FANUC Controllers with License-Free OEE Analytics

Scalable monitoring platform for 10+ CNC machines, eliminating FANUC proprietary licensing costs.

Executive Summary

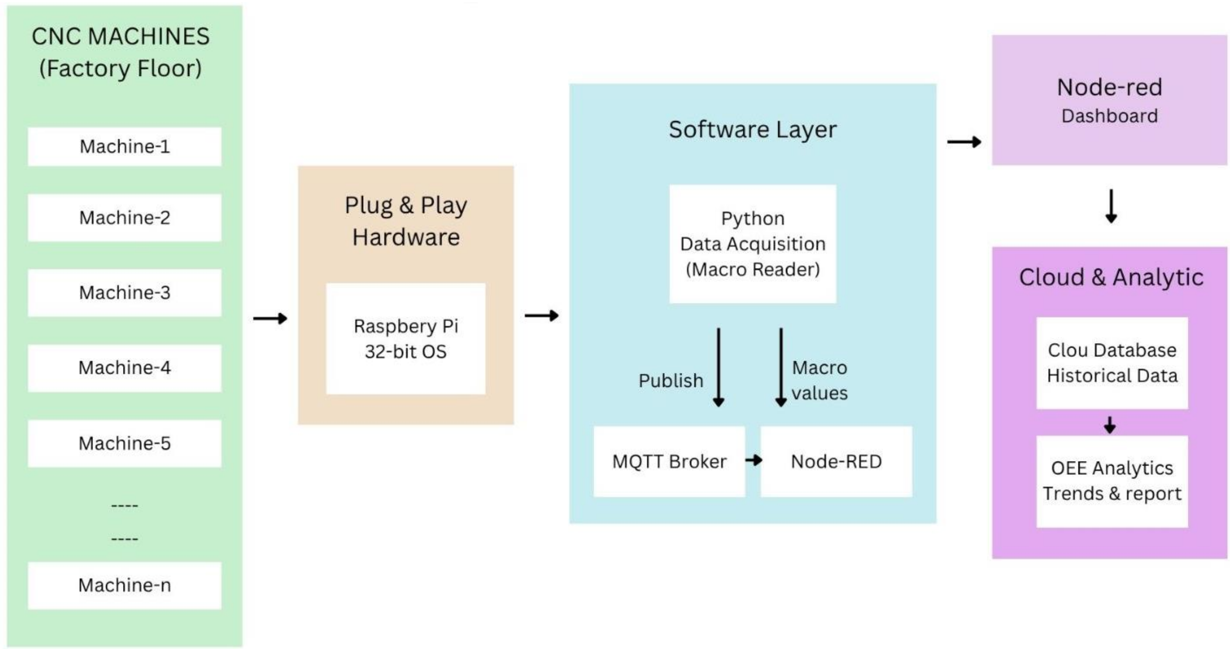
Problem:
Industrial CNC machines (especially FANUC) require expensive licenses (e.g., FANUC FOCAS, MT-Linki) to extract real-time OEE data. Small and medium manufacturers cannot justify such costs.

Solution:
We developed a dual-layer approach that eliminates licensing fees:

1. Embedded G-code macros inside the CNC program store OEE metrics (parts, run time, cycle time) in standard macro variables (#600–#612).
2. A Python script reads these variables over Ethernet using the free FOCAS library (no extra license needed for reading macros).

The result is a complete, real-time OEE monitoring system with no per-machine licensing costs, deployed on 8+ machines at CMTI's Smart Manufacturing Demo Cell.

System Architecture



1. Ethernet connection between PC (or Raspberry Pi) and CNC using FANUC's FOCAS library over TCP/IP.
2. No extra hardware beyond a standard Ethernet cable.
3. Data flow: Every second, the Python script requests macro values → calculates OEE → publishes to Node-RED → displays real-time dashboard.

Data Acquisition Strategy

G-code Macro Variables are stored inside the CNC controller using common macro variables.

Note: This part requires no license and works on any FANUC (or compatible) CNC.

Macro	Description	Update Method
#610	Planned quantity	Set manually per shift
#611	Ideal cycle time (sec)	Programmed constant
#612	Job ID	Programmed per job
#600	Total produced	Incremented every cycle
#601	Good parts	Incremented if no reject
#602	Reject parts	Incremented on reject
#603	Actual run time (ms)	Accumulated per cycle
#604	Actual cycle time (ms)	Calculated each cycle

Ethernet-Based Reading via FANUC FOCAS

1. The Python script connects to the CNC via Ethernet using cnc_allclibhndl3 (FOCAS library).
2. No additional license – the FOCAS DLL is freely distributable and allows reading macro variables, status, positions, etc.
3. The script reads macro variables #600–#612 once per second.

G-Code Macros for OEE

1. O9000 – Initialisation & Cycle Start

```
O9000 (OEE CONTROL MACRO)
(---------------- INITIALIZATION ----------------)
IF [#699 EQ #612] GOTO 100
#600 = 0    (Total produced)
#601 = 0    (Good parts)
#602 = 0    (Reject parts)
#603 = 0    (Run time)
#604 = 0    (Cycle time)
#699 = #612

N100
(---------------- CYCLE START ----------------)
#650 = 0    (good parts flag)
#651 = 0    (incomplete execution flag)
#700 = #3001    (start milliseconds)
M99
```

2. O9001 – Cycle End & Part Counting

```
O9001 (OEE END MACRO)
#701 = #3001
#604 = #701 - #700
#603 = #603 + #604
#651 = 1
#600 = #600 + 1
```

```
IF [#650 EQ 1] GOTO 10
IF [#651 EQ 0] GOTO 10
#601 = #601 + 1
GOTO 20
N10 #602 = #602 + 1
N20 M99
```

3. Production Program Example (O1000)

```
O1000 (PRODUCTION PROGRAM – JOB A)
#610 = 100    (Planned qty)
#611 = 60     (Ideal CT)
#612 = 101    (Job ID)

M98 P9000    (OEE START)

(=============== MACHINING ================)

M5
M98 P9001    (OEE END)
M30
```

Python Data Acquisition & OEE Calculation

1. Connects to CNC via Ethernet using cnc_allclibhndl3 (FOCAS).
2. Reads macro variables #600–#612 using cnc_rdmacro.
3. Reads machine status (cnc_statinfo) to detect alarms, running state, etc.
4. Calculates OEE (Availability, Performance, Quality).
5. Publishes all values to an MQTT broker for Node-RED.

Key Code Snippet

```
# Connect to CNC
handle = c_ushort()
fwlib.cnc_allclibhndl3(b"x.x.x.x", xxxx, 10, byref(handle))

# Read macro #600
macro_600 = ODBMACRO()
fwlib.cnc_rdmacro(handle.value, 600, ctypes.sizeof(ODBMACRO), byref(macro_600))
total_parts = macro_600.data / (10 ** macro_600.dec)

# OEE calculation (simplified)
planned_time_sec = planned_qty * ideal_ct
availability = run_time_sec / planned_time_sec
performance = (total_parts * ideal_ct) / run_time_sec
quality = good_parts / total_parts
oee = availability * performance * quality * 100
```

Conclusion

We have demonstrated a **complete, cost-effective, license-free OEE monitoring ecosystem** for FANUC CNCs using:

1. Embedded G-code macros (O9000/O9001) – no per-machine fee.
2. Ethernet-based data reading via FOCAS – no extra license.
3. A user-friendly Node-RED tool to automatically inject OEE macros into any G-code program.
4. Open-source components (Python, MQTT, Node-RED).

The system is fully operational at CMTI and can be replicated by any factory with macro-enabled CNCs.